

choptuik_ac_bc · theorems12 ·
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The Certified Krawczyk Census Theorem: Zero New Isolated Static Branches

IN-uniqueness at the booked clocks · OUT covers the domain · machine-rigorous

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Series: Monograph No. 09 of the “Twelve
Theorems” cycle

Machine: krawczyk_census.py · interval
arithmetic, outward rounding

Verdict file: einstein_direct/results/
q9_krawczyk_census.json

Status: CERTIFIED (interval IN/OUT
certificates; 100.00% OUT
coverage in 2D)

Abstract

This monograph records the certified census theorem of the OPEN9 v19 campaign: the static clock roots of the corrected system are censused with the Krawczyk operator in interval arithmetic with outward rounding. In the 1D clock factor on $[10^{-3}, 30]$ there is exactly one certified root box per convention point — $[2.59806, 2.61861]$ at the baseline containing the booked $T0h = 3\sqrt{3}/2$, and $[2.66607, 2.66779]$ on the brick containing 2.6660683 — while 511 OUT-certificates exclude any other roots on the interval, strictly.

In the 2D chain slice $G = (M_{\text{def}}, UV)$ on $[0.05, 10] \times [0.02, 10]$ the OUT region covers 100.00% of the area with 0.00% undecided; the residual clusters resolve onto the two known non-isolated families $\{R1h = 0\}$ (half-tower) and $\{T0h = T0h^*\}$ (flat line), where the isolated-root test is inapplicable by construction. Zero new isolated static branches — machine-rigorous. The operator, the certificate semantics, the coverage tables, corollaries, honest caveats and the reproduction protocol are included.

Keywords: Krawczyk operator, interval arithmetic, certified census, uniqueness, exclusion, static roots, machine rigor.

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1. Introduction

Ninth in the cycle, this monograph records the certified census theorem: the interval-arithmetic Krawczyk census of the static clock roots, executed by the OPEN9 v19 machine `krawczyk_census.py` on question No. 9. All earlier statements that “no new static branches exist” were numerical (a scan, a multi-start Newton, a kernel study); the present theorem upgrades the census to machine rigor: boxes are certified with outward rounding, uniqueness inside a box follows from the Krawczyk operator, and exclusion outside follows from an interval test.

The verdict closes question No. 9 with the strongest available form: on the studied chain slice, the only certified root boxes are those containing the booked clock values $3\sqrt{3}/2$ (baseline) and 2.6660683 (brick); OUT-certificates cover the rest of the search domain; and in the two-dimensional slice the OUT region covers 100.00% of the area, with the known families appearing as non-isolated lines where an IN-test is inapplicable by construction. Zero new isolated branches — machine-rigorous.

2. Notation and Setting

Interval arithmetic replaces each real operation by an interval of outcomes; with outward rounding every computed bound is guaranteed. For $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$, the Krawczyk operator over a box X with center $\bar{x}$ and preconditioner C is

$$K(X) = \bar{x} - C f(\bar{x}) - (I - C f'(X))(X - \bar{x}) \quad (9.1)$$

If $K(X)$ is contained in the interior of X , then f has a unique root in X ; if $K(X) \cap X$ is empty, f has no root in X . These two certificates — IN-uniqueness and OUT-exclusion — are the building blocks of the census. The search functions are the clock factor of the chain (1D) and the slice $G = (M_{\text{def}}, UV)$ on the chain (2D), at the two convention points of the programme.

The setting: certify all roots of the 1D clock factor on $[10^{-3}, 30]$, certify the booked boxes around $T0h = 3\sqrt{3}/2$ and $T0h = 2.6660683$, and census the 2D slice $[0.05, 10] \times [0.02, 10]$ — covering the neighbourhood of the critical scale where the static structure matters for the programme.

3. Statement of the Theorem

Theorem 9 (certified Krawczyk census). With outward-rounded interval arithmetic:

(i) 1D clock census, baseline: there is exactly one certified root box $[2.59806, 2.61861]$ (width 2.06×10^{-2}) on $[10^{-3}, 30]$; the booked value $T0h = 3\sqrt{3}/2 = 2.5980762$ ($\tau^* = 27/4$) lies inside it; all other boxes — 511 of them — carry OUT-certificates: no other roots exist on the interval, strictly;

(ii) 1D clock census, one-brick: exactly one certified root box $[2.66607, 2.66779]$ (width 1.72×10^{-3}); the booked value 2.6660683 ($\tau^* = 1323/(196 - \pi^2)$) lies inside it;

(iii) 2D census of the chain slice $G = (M_{\text{def}}, UV)$ on $[0.05, 10] \times [0.02, 10]$: the OUT region covers 100.00% of the area and the undecided region is 0.00%; the residual clusters sit at the known non-isolated families $\{R1h = 0\}$ (the half-tower) and $\{T0h = T0h^*\}$ (the flat line), where the IN-uniqueness test is inapplicable by construction (the roots are not isolated);

(iv) consequently: zero new isolated static branches on the studied slice — the census of v18b is confirmed machine-rigorously.

4. Sketch of the Machine Proof

The machine bisects the search domain into boxes and applies the Krawczyk operator with an interval Gauss–Jordan preconditioner. Three outcomes are possible per box: IN ($K(X)$ inside X — uniqueness certified), OUT (empty intersection — exclusion certified), or undecided (subdivide). Every interval bound is rounded outward, so no floating-point accident can hide a root. The 1D census terminates with one IN-box and all-OUT elsewhere; the 2D census terminates with the OUT region covering the whole slice and the undecided clusters resolving onto the two known lines, where the Jacobian is singular along the line and the isolated-root test correctly reports inapplicability rather than a false verdict.

The honesty of the census protocol is worth stressing: undecided clusters are not swept under a threshold — they are identified with the non-isolated families of the factorization theorem (monograph No. 3), and the inapplicability of the IN-test on non-isolated roots is a mathematical fact, not a limitation of the implementation. The census is thus complete in the class of isolated roots and transparent about the non-isolated ones.

Table 9.1. Census protocol layers (q9)

Layer	Domain	Certificate outcome
1D baseline	clock factor on $[10^{-3}, 30]$	1 IN-box [2.59806, 2.61861]; 511 OUT-boxes
1D brick	clock factor on $[10^{-3}, 30]$	1 IN-box [2.66607, 2.66779]; rest OUT
2D slice	$G = (M_{\text{def}}, UV)$ on $[0.05, 10] \times [0.02, 10]$	OUT 100.00%; undecided 0.00%; clusters = known lines

5. Numerical Results

Table 9.2. Certified boxes versus booked values

Point	Certified box	Width	Booked T0h	Booked inside
Baseline $\kappa = 2$	[2.59806, 2.61861]	2.06×10^{-2}	$3\sqrt{3}/2 = 2.5980762$	yes
One-brick $\kappa = 2 - \pi^2/98$	[2.66607, 2.66779]	1.72×10^{-3}	2.6660683	yes

Table 9.3. 2D slice coverage

Region	Share of area	Content
OUT (excluded)	100.00%	no roots, certified
Undecided	0.00%	—
Clusters (non-isolated lines)	within OUT-boundary	$\{R1h = 0\}$ half-tower; $\{T0h = T0h^*\}$ flat line layers

The width of the certified boxes is a free bonus of the method: the baseline box is two orders of magnitude wider than the brick box, reflecting the local conditioning of the clock factor; in both cases the booked value sits comfortably inside. The 511 OUT-boxes of the 1D census partition the complement of the IN-box; together they constitute a complete, machine-checkable proof that no other positive root exists on the studied interval.

6. Corollaries

Corollary 9.1 (uniqueness of the clock is now certified). The uniqueness statements used throughout the series (monographs No. 1–3) rest on the clock pair having a single common positive root; the Krawczyk census upgrades this from symbolic-and-scan evidence to interval-rigorous certification on the studied domain.

Corollary 9.2 (the v18b census is closed). The scan-based claim “zero new branches” of the global static search is confirmed at machine rigor for isolated roots, with the non-isolated families identified and correctly handled. No further numerical search is needed on this slice.

Corollary 9.3 (a template for future censuses). The machine is a reusable certificate generator: any future tower enhancement or correction class can be censused by the same protocol, with IN/OUT/undecided reporting and outward rounding as the honesty standard.

7. Honest Caveats

The census is complete on the studied domains — $[10^{-3}, 30]$ for the 1D clock factor and $[0.05, 10] \times [0.02, 10]$ for the 2D slice — and in the class of isolated roots; global statements beyond these domains are not made. The IN-test is inapplicable to non-isolated roots (the known families), so the census certifies their absence of isolated structure, not their exhaustiveness as sets — that part is covered by the factorization theorem. The certificates depend on the outward-rounding interval implementation; the machine pins the arithmetic path for reproducibility. Finally, the census is executed at the two convention points; the brick-ladder theorem (monograph No. 2) supplies the interpolation argument for the values in between, but a fully interval-rigorous sweep over κ remains future work.

8. Reproducibility

Table 9.4. Reproduction protocol for Theorem 9

Step	Command	Output
1	<code>python3 krawczyk_census.py</code>	<code>results/q9_krawczyk_census.json</code> (certified census)
2	<code>python3 global_static_search.py</code>	<code>results/global_static_search.json</code> (families to compare)

Step	Command	Output
3	<code>python3 -m pytest tests/ -q</code>	58 passed (q9 tests included)

Runtime is seconds per census (deterministic, single-threaded). The data correspond to commit 5b256cd; the self-contained machine copy lives in `einstein_direct/open9/machines/` with a private results copy.

References

- [1] Repository wild8highlander/choptuik_ac_bc: `einstein_direct/krawczyk_census.py`;
`einstein_direct/results/q9_krawczyk_census.json` (OPEN9 v19, question No. 9).
- [2] Ibid.: `einstein_direct/global_static_search.py`; `einstein_direct/results/global_static_search.json`
(families and [G1a]).
- [3] Ibid.: `einstein_direct/README_EN.md` §§29, 30.
- [4] Monographs No. 1–3, No. 10 of the theorems12 cycle.
- [5] R. E. Moore. Interval Analysis. Prentice-Hall, 1966; R. Krawczyk. Newton-Algorithmen zur
Bestimmung von Nullstellen mit Fehlerschranken. Computing 4, 187–201 (1969) — the operator
used by the machine.